1 DH description

The six revolute joints are described using the *standard* DH convention. Frame $\{0\}$ coincides with `base_link`, and $q_i = 0$ denotes the URDF home configuration. The endpoint E is `link_6`. For each step i, the transform from frame $\{i-1\}$ to frame $\{i\}$ is

$$^{i-1}T_i = R_z(\theta_i)\, T_z(d_i)\, T_x(a_i)\, R_x(\alpha_i), \qquad \theta_i = q_i + \theta_i^0. \tag{1}$$

Joint i rotates about z_{i-1}. For $i < 6$, the new axis z_i lies on joint $i+1$; the terminal axis z_6 is chosen parallel to z_5. DH frame numbers therefore differ from physical joint numbers.

How to read one step. Choose x_i along a common normal to z_{i-1} and z_i. Its intersection with z_i defines O_i. If two distinct axes intersect, their common-normal length is zero and x_i is perpendicular to both axes through their intersection. For coincident axes, the origin and transverse x axis are free; here the last origin is the endpoint. Set $y_i = z_i \times x_i$ to complete a right-handed frame. The four parameters then have the following geometric meaning:

Parameter	Measured about / along	Construction from $\{i-1\}$ to $\{i\}$
θ_i	about z_{i-1}	Turn x_{i-1} towards x_i.
d_i	along z_{i-1}	Move from O_{i-1} to the foot of the common normal.
a_i	along x_i	Follow the common normal to O_i.
α_i	about x_i	Turn z_{i-1} towards z_i.

Translations are signed along the indicated positive axes; rotations follow the right-hand rule. Figure 1 shows the construction at $q = 0$.

Table 1: Six DH steps. Lengths are in metres; angular constants are in radians.

i	d_i	θ_i	α_i	a_i	Geometric construction at home
1	1	q_1	$-\pi/3$	0.8	Rise $0.5 + 0.5$ from the base, then move 0.8 along x_1 to joint 2. Its axis gives a $-60°$ twist.
2	1.25	q_2	$+\pi/2$	2	Move 1.25 along joint 2's axis and 2 along x_2 to joint 3; twist by $+90°$.
3	0.75	$q_3 + \pi/2$	$+\pi/2$	0	Extend joints 3 and 4's axes to their intersection O_3. Turn x_2 by $+90°$ to select x_3; no common-normal translation is needed.
4	2.5	q_4	$-\pi/2$	0	From O_3, follow joint 4's axis by $1.5 + 1$ to the wrist centre O_4; twist by $-90°$.
5	0	q_5	$+\pi/2$	0	Joints 5 and 6's axes meet at the wrist centre. Keep $O_5 = O_4$ and twist by $+90°$.
6	1.5	q_6	0	0	Follow joint 6's axis from the wrist centre to the endpoint. Choose $z_6 = z_5$ and no fixed twist.

Frame origins versus joint centres. A DH origin lies on the selected joint axis but need not coincide with a URDF joint origin. In particular, O_3 lies 1.5 m before the physical centre of joint 4. This distance enters d_4, giving $d_4 = 1.5 + 1 = 2.5$ m, while $a_3 = 0$. Similarly, $d_5 = 0$ even though the centres of joints 5 and 6 are 1.5 m apart: the two axes already intersect at $O_4 = O_5$.

Endpoint alignment. Frame $\{6\}$ and the URDF endpoint have the same origin but different axes: $x_E = z_6$, $y_E = x_6$ and $z_E = y_6$. Hence the forward kinematics uses the fixed alignment on the right:

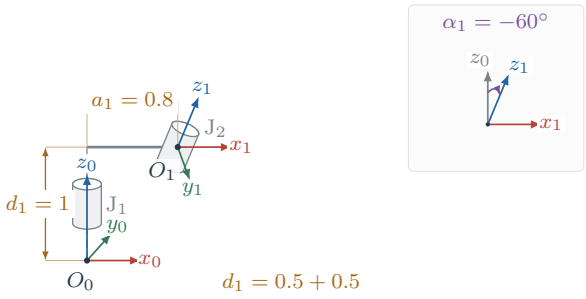
$$^0T_E = \left(\prod_{i=1}^{6} {}^{i-1}T_i\right) {}^6T_E, \qquad {}^6T_E = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{2}$$

The six frame-to-frame constructions

Home configuration ($q = 0$), with a fixed viewing direction throughout.

1 Base to shoulder: $\{0\} \to \{1\}$

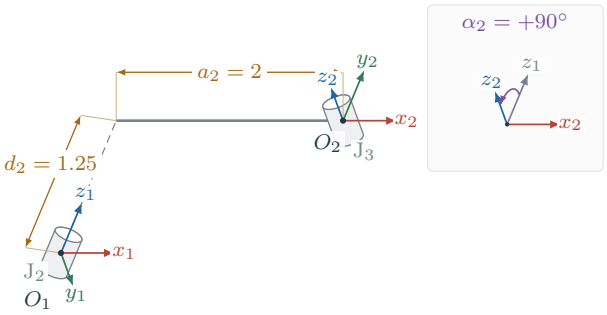
$d_1 = 1, \quad \theta_1 = q_1, \quad \alpha_1 = -60°, \quad a_1 = 0.8$



The full base height is 1 m. The 30° URDF roll of joint 2 gives a $-60°$ DH twist because its local axis is y.

2 Shoulder to elbow: $\{1\} \to \{2\}$

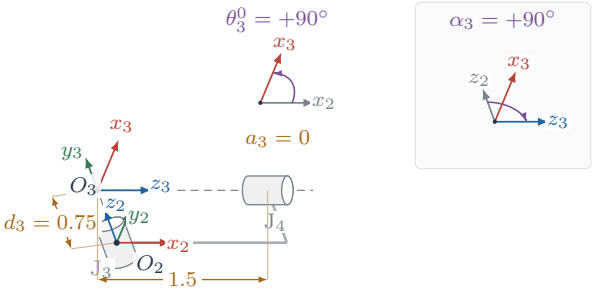
$d_2 = 1.25, \quad \theta_2 = q_2, \quad \alpha_2 = 90°, \quad a_2 = 2$



The x axes remain parallel. The offset along z_1 and the common normal along x_2 locate O_2 on joint 3.

3 Axis intersection: $\{2\} \to \{3\}$

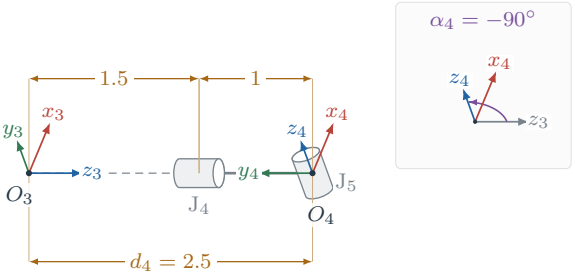
$d_3 = 0.75, \quad \theta_3 = q_3 + 90°, \quad \alpha_3 = 90°, \quad a_3 = 0$



O_3 is the intersection of the extended axes. Choosing $x_3 = y_2$ explains the fixed $+90°$ offset in θ_3.

4 Reach the wrist: $\{3\} \to \{4\}$

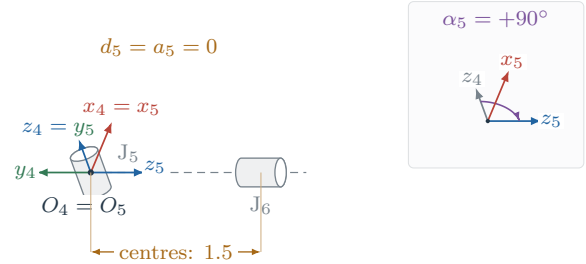
$d_4 = 2.5, \quad \theta_4 = q_4, \quad \alpha_4 = -90°, \quad a_4 = 0$



The distance starts at O_3, rather than the joint-4 cylinder. The two physical offsets add: $d_4 = 1.5 + 1$.

5 Coincident origins: $\{4\} \to \{5\}$

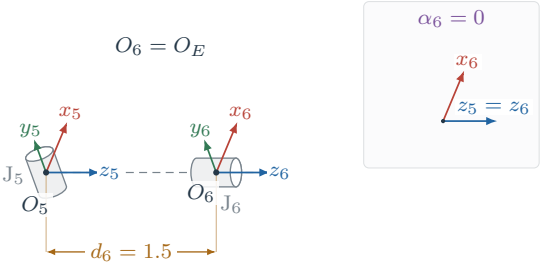
$d_5 = 0, \quad \theta_5 = q_5, \quad \alpha_5 = 90°, \quad a_5 = 0$



The origins coincide, although the joint centres do not. Here $x_4 = x_5$ and the extended joint-6 axis passes through the wrist centre.

6 Wrist to endpoint: $\{5\} \to \{6\}$

$d_6 = 1.5, \quad \theta_6 = q_6, \quad \alpha_6 = 0, \quad a_6 = 0$



At home the two frames have parallel axes. $O_6 = O_E$; the fixed endpoint alignment changes orientation only.

Figure 1: Standard DH frame assignment. Cylinders denote physical revolute joints J$_i$; dots mark DH origins O_i. x, y and z axes retain their colours. Dashed lines extend joint axes; gold dimensions locate the origins. Angle sketches show θ_3^0 and the signed twist $z_{i-1} \to z_i$ about $+x_i$ (with the first direction translated to the same point). Cylinder orientation, size and viewing direction are reused between panels; drawings are schematic.